

Slide	Notes
C O D E Lesson 1: Talking to Machines Problem Solving with AI	
Warm Up	Warm Up (10 mins)
Problem Solving with AI Talking to Machines – Warm Up ✔ Do This: Start with the following phrase: A mysterious talking cat started showing up to math class. In your group, take turns building on the conversation by starting your next sentence with "Yes, and . . ."	Yes, and... ■ ■ Do This: Group students in pairs or groups of three and read aloud the sentence on the board. Have students take turns one-by-one building on the conversation by starting their next sentence with "Yes, and.." Teaching Tip For example: Opening Statement: "A mysterious talking cat started showing up to math class." Student 1: "Yes, and he said his name is Whiskers." Student 2: "Yes, and he happens to be very good at multiplication." Student 1: "Yes, and he has a sister who started showing up to English class." (Optional) If you have time to play multiple rounds of the game, here are some additional opening statements: "The class down the hall just discovered a hidden tunnel beneath the school." "The principal announced that prom will be held on a spaceship this year." "Every locker in the main hallway suddenly started playing music."
Problem Solving with AI Talking to Machines – Warm Up 💬 Discuss: Summarize the story you came up with. What were the last two sentences of your conversation?	■ Discuss: Summarize the story you came up with. What were the last two sentences of your conversation? Have several groups summarize where their conversation went. Then have them share out their last sentence or two that they came up with. ■ Discussion Goal: Students should recognize that all groups started with the same sentence starter but may have ended up with very different conversations. Have students note any patterns between the stories as well as any differences. This game leads into the main activity by explaining how AI generates content.

AI Connections

A mysterious talking cat started showing up to math class.

Yes, and...

Yes, and...

Yes, and...

Everyone thought it was a prank

She always sat in the front row

The teacher tried to shoo him away

■ Do This: Have students read the different sentences on this slide.

■ Say: Generative AI is like a “Yes, and” machine. It builds on existing sentences by finding a word or phrase that would make sense to come next.

Lesson Objectives

By the end of this lesson, you will be able to . . .

- Explain that AI models use probability and statistics to generate responses.
- Experiment with different prompts to observe how AI's responses change.
- Analyze patterns in AI-generated responses to explain how input prompts influence outputs.
- Refine AI-generated outputs by iterating on prompts and recognizing patterns in responses.

■ Do This: Introduce the lesson objectives.

Question of the Day

How can I use AI effectively?

■ Do This: Introduce the Question of the Day.

Activity (30 mins)

Activity

■ Video: Problem Solving with AI



■ Video: Show Problem Solving with AI video.

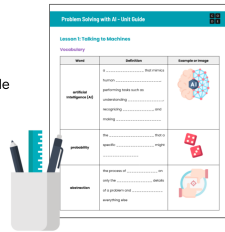
Teaching Tip

To encourage active engagement and reflection, use one or more of the strategies discussed in the Guide to Curriculum Videos.

Unit Guide

You should have:

- Problem Solving with AI Unit Guide
- pen or pencil

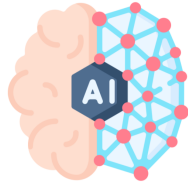


■ Distribute: Give each student a copy of the Problem Solving with AI Unit Guide.

What is AI?

Artificial intelligence (AI)

is a technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions.



Unit Guide

AI Predictions (10 mins)

■ Do This: Define artificial intelligence.

Discuss:

Predict the word that you think will probably come next in the following sentence:

The dog barked at the . . .

■ Discuss: Predict the word that you think will probably come next in the following sentence: The dog barked at the...

Ask students to predict the word that they think will probably come next (i.e. the most common word) and provide a reason for why they chose that word. Have all students raise their hand when they think of an answer. Start calling on students and have other students put down their hand if someone is called on who has the same answer.

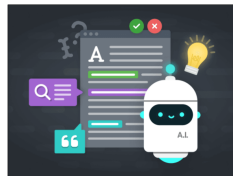
■ Discussion Goal: Students will see that there are several common possibilities that are chosen. Common words might include squirrel, mailman, stranger, etc. However, there will also be outliers that might not be as common (e.g. my aunt Lucy). Having students recognize that there are more and less common responses is a first step towards understanding how probability works while making this sort of prediction.

AI Predictions

Lesson 1

Do This:

1. Go to Lesson 1, Level 1.
2. Ask AI to generate the next word in the sentence: *The dog barked at the . . .*



■ Do This: Model the AI Chat in Lesson 1, Level 1 for students.

■ Transition: Have students experiment with using the AI Chat to complete the sentence.

Teaching Tip

Have AI generate several outcomes to complete the sentence. You can keep prompting AI Chat using the same prompt or follow up by prompting it to "generate three more answers". You may also want to have students try out the prompt on their devices and compare responses.

Discuss:

How does AI's output compare to the word you predicted?

■ Discuss: How does AI's output compare to the word you predicted?

■ Discussion Goal: Students should start to see how AI selects words based on probability. It tends to choose the most likely next word, which often mirrors our own prediction of the most-likely next word.

What is Probability?

When making predictions, **probability** takes into account the likelihood that a specific outcome might occur.



Unit Guide

■ ■ Do This: Define probability.

Successful Predictions

Complete the following sentences:

"He heard a knock at the..."

"Thank..."

"As the bell rang, they put down their..."



"He heard a knock at the... **door**"

"Thank... **you for your help**"

"As the bell rang, they put down their... **pencils**"

■ ■ Do This: Read through this scenario of probability-driven predictions that make sense and are successful. Make sure students understand that the grey box represents the user input, and the blue box below represents the AI Chat response.

■ ■ Say: AI uses the computational skill of pattern recognition to generate its most likely response. For example, it reads over thousands of sentences that contain the different phrases above (e.g. "heard the knock at the...") and starts to recognize a pattern.

Failed Predictions

Probability-driven predictions can fail because they don't rely on meaning.

Draw "the keys are on the table".



■ ■ Do This: Examine this scenario where probability-driven predictions fail because they don't rely on meaning.

Why Did the AI Fail?

The AI focused on the words "keys", "on", and "table" and abstracted away unnecessary details of the prompt.

Draw "the **keys** are **on** the **table**".



■ ■ Do This: Explain that AI focused on the words "key", "on", and "table" and abstracted away unnecessary details of the prompt.

■ ■ Say: Since AI doesn't understand meaning, it drew a single key resting on the table in an unrealistic way. It prioritized the literal meaning of the essential details of the sentence rather than considering the sentence as a whole.

What is Abstraction?

Abstraction is the process of focusing on only the essential details of a problem and ignoring everything else.



Unit Guide

■ ■ Do This: Define abstraction.

Teaching Tip

Abstraction is one of the four pillars of computational thinking– a framework designed for solving problems. This concept will be repeated throughout AIF units.

What is a Prompt?

A **prompt** is a question, instruction, scenario, or statement provided by the user to guide the AI's response.


[Unit Guide](#)
Prompting AI (20 mins)

■ Do This: Define prompt.

Prompting AI**Lesson 1**

○ 2 ○ ○ ○ ○

Do This:

1. Go to Lesson 1, Level 2.
2. Complete Levels 2 through 4 by copying and pasting the prompts into AI Chat.
3. Read the responses that it gives!



■ Transition: Direct students to Lesson 1, Level 2 to complete Levels 2 through 4.

Level 2: Students input "Give me 3 gift ideas" into AI Chat.

Level 3: Students input a more specific prompt that defines the person who is getting the gift.

Level 4: Students add an image to accompany the specific prompt that they reuse from Level 3.

Video:

What tips might you give to someone else who is looking to improve their prompts?



■ Video: Show the "Plan a Perfect Trip with ChatGPT" video.

Questions to consider with the video:

What tips might you give to someone else who is looking to improve their prompts?

■ Discussion Goal: Students might point out that they should include both information that they are interested in getting, but also information that they are not. They might also mention that including demographic information and specific numbers are helpful for more useful responses.

Teaching Tip

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

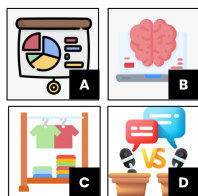
To encourage active engagement and reflection, use one or more of the strategies discussed in the Guide to Curriculum Videos.

Prompting Practice**Lesson 1**

○ ○ ○ 5 ○

Do This:

1. Go to Lesson 1, Level 5.
2. Choose one of the chatbots configured to assist with a specific task.
3. Follow the instructions for how to interact with the chatbot.



■ Transition: Direct students to Lesson 1, Level 5. Students choose one of the AI Chat levels that is configured for a specific task. They start by giving the AI Chat a generic prompt first (such as "Help me prepare for my class presentation."), then make it more specific, then make it more specific and add an image. Students should update their prompt or interact with the AI Chat in at least three back and forth interactions.

Teaching Tip

This is a choice level, so students only need to complete one of the levels.

However, if students finish early or are looking for an additional challenge, they can choose Option D: Debate an AI Partner level. This option is more challenging because it requires students to engage in a back-and-forth exchange with AI, refine their arguments, and adapt their reasoning based on AI's responses. Unlike other choices that focus on planning tasks, this level encourages students to critically analyze AI-generated arguments and practice constructing persuasive responses.

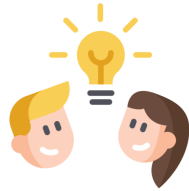
Check Your Understanding

Lesson 1

○○○○○6

Do This:

1. Go to Lesson 1, Level 6.
2. Respond to the question.



■■■ Do This: Direct students to Lesson 1, Level 6 to complete the formative assessment. Display students' anonymous responses.

Assessment Opportunity

This level can be used as a formative assessment to check student understanding of the lesson objectives. Here are things to look for and actions you can take:

If students are only describing what happened rather than why it happened, prompt them to revisit earlier lesson segments on probability-driven outputs. Consider a quick group discussion where students share examples of prompts that worked well and why those prompts influenced AI's response.

If students seem uncertain about constructing high-quality prompts, have them re-examine Level 5 where best prompting strategies are summarized. They can also do a brief pair activity—each student provides a vague prompt, then a specific prompt, and compares the results.

If many students need more help, provide short demos or additional videos showing how multimodal inputs can change the AI's outputs. Or point them to the optional video about planning a family trip to emphasize how refining prompts improves results. Encourage a follow-up activity where students generate multiple answers to the same prompt, then analyze and explain what changed.

Wrap Up

Wrap Up (5 mins)

Today, you learned about . . .

- How AI models use probability and statistics to generate responses.
- How different prompts impact AI's responses.
- Patterns in AI-generated responses and how prompts influence outputs.
- Refining AI-generated outputs by iterating on prompts.

■■■ Do This: Review the lesson objectives.

Question of the Day

How can I use AI effectively?

■■■ Do This: Review the Question of the Day.

Rapid Roundtable

In a circle or small group, answer the question of the day using one sentence or key phrase:

Question of the Day: How can I use AI effectively?

How many unique answers can you come up with?



Rapid Roundtable (5 mins)

■ ■ Do This: Have students sit in a circle or small groups. Pose the Question of the Day to students. One by one, they quickly state their answer to the Question of the Day. Responses should be brief (one sentence or key phrase). They should see how many unique answers they can come up with.

Teaching Tip

If students are struggling to answer the Question of the Day, have them think about which prompting strategies got them the most helpful AI Chat outputs. For example, ask students what details they included in their prompts or how they selected an image to add to their prompt.



Key Vocabulary

- **artificial intelligence (AI):** a technology that mimics human intelligence, performing tasks such as understanding language, recognizing patterns, and making decisions
- **probability:** the likelihood that a specific outcome might occur
- **abstraction:** the process of focusing on only the essential details of a problem and ignoring everything else
- **prompt:** a question, instruction, scenario, or statement provided by the user to guide the AI's response



■ ■ Do This: Review the Key Vocabulary.